

Graph Analysis

Translating Complex Relationships into Actionable Knowledge

Version 1.0

Alexandre Soares



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por

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Source Code

All source programs and data used for the preparation of this study, as well as the PDF of this material, are available for download at the GitHub repository, at the address: <https://github.com/Alxsoa/Artigos>.

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1

Introduction

1.1. Understanding Networks Through Data Exploration

Data exploration in a graph, known as EDA (Exploratory Data Analysis), is an essential step to understanding the structure, patterns, and hidden behaviors within a complex network. Unlike tabular data, where relationships between observations are implicit, in a graph, connections are explicit and carry much of the relevant information.

Conducting exploratory analysis allows us to identify how vertices relate to each other, which nodes are most influential, and how the network organizes itself in terms of communities or information flows.

For example, calculating metrics such as degree, centrality, clustering, and diameter helps to understand whether the network is dense, hierarchical, or decentralized. Betweenness centrality highlights nodes that act as critical "bridges" and identifies strategic elements for communication between groups. Connectivity analysis reveals whether the network is robust to failures, while component detection identifies possible isolated sub-networks.

Furthermore, exploratory data analysis (EDA) on graphs is fundamental to supporting practical decisions. In social networks, it indicates influencers and information spreaders; in electrical or logistics networks, it points out structural vulnerabilities; and in biological systems, it shows crucial interactions between molecules or organisms.

Therefore, exploring the data of a graph is not just a technical matter but a way to translate complex relationships into actionable knowledge, allowing us to better understand the system, anticipate failures, and optimize its overall efficiency.

2

Data Used for the Study

2.1. Zachary's Karate Club Graph

The graph presented in Figure-2.1 is a representation of Zachary's Karate Club Graph, one of the most classic and widely used datasets in social network analysis and graph theory. This graph, available directly in the NetworkX library through the command `nx.karate_club_graph()`, models the friendship relationships between 34 members of a karate club at an American university, studied by Wayne Zachary between 1970 and 1972.

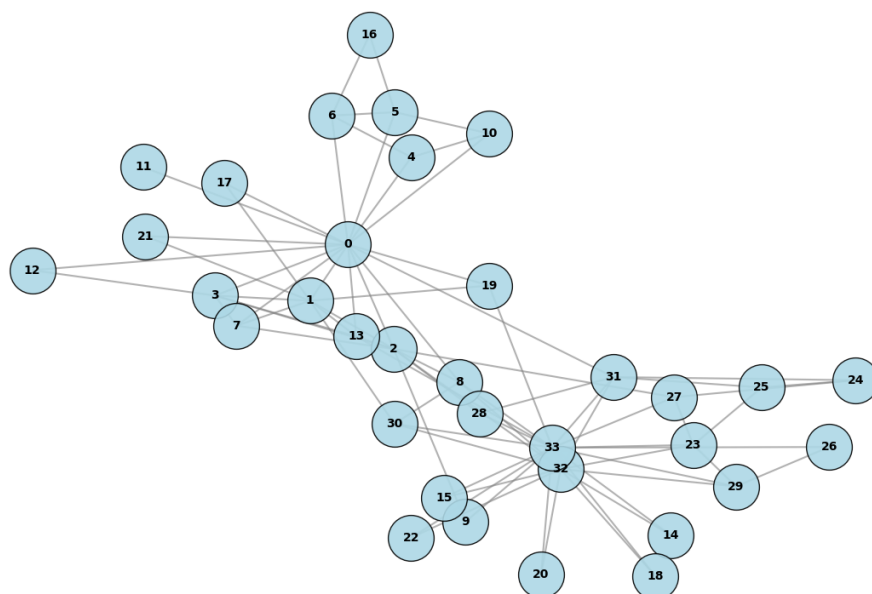


Figure 2.1: Base Graph Used in EDA

Each node (vertex) represents a club member, and each edge indicates a friendship or social interaction relationship between two members. The connections were mapped through direct observation over two years. The graph is undirected, as friendship relationships are considered mutual; furthermore, it is unweighted since all connections have the same importance.

Zachary's study became famous because the club eventually split into two distinct groups due to a conflict between the instructor (node 0) and the administrative president (node 33). This real-world division allowed for a comparison of mathematical methods of community detection with observed social behavior. In the graph, node 0 appears as one of the most central, with several direct connections; this reflects his leadership role as an instructor. Meanwhile, node 33 also exhibits high centrality, as he leads the group that separated after the conflict.

In terms of structure, this graph contains 34 nodes and 78 edges, with a complex topology that includes highly connected nodes (hubs) and several triangles formed by friendship subgroups. Therefore, it is widely used to study properties such as Degree Centrality, which identifies the most popular members (such as nodes 0 and 33); Betweenness Centrality reveals who acts as a bridge between distinct social groups; Clustering Coefficient measures how much an individual's friends are also friends with each other; Community Detection allows for observing the natural division of groups, reflecting the historical event of separation.

Thus, the Karate Club Graph is a fundamental tool for understanding how social networks form, evolve, and separate. It clearly illustrates how the mathematical properties of a graph can describe real and complex human behaviors and is still used today in studies on social influence, group dynamics, and complex network theory.

3

Node Degree Distribution

3.1. Understanding the Structure and Connections of a Network

Figures-3.1 and 3.2 analyze the degree distribution of nodes in Zachary's Karate Club Graph, allowing us to understand the structure and how connections are distributed among network members.

Figure-3.1, called "Degree Distribution (Log-Log Scale)," presents on the horizontal axis (k) the node degree, that is, the number of connections each vertex has, and on the vertical axis ($P(k)$), the probability of occurrence of these degrees in the network.

When represented on a logarithmic scale, the graph shows the decay pattern of the frequency of nodes with a high degree. It is observed that most nodes have few links, while few nodes concentrate a large number of connections.

This behavior is characteristic of complex networks with a heterogeneous structure, in which few elements play a central role, functioning as highly connected nodes (hubs). In the club context, this means that only some members, such as the instructor and the president, maintain direct relationships with many other participants, while most members connect with few colleagues.

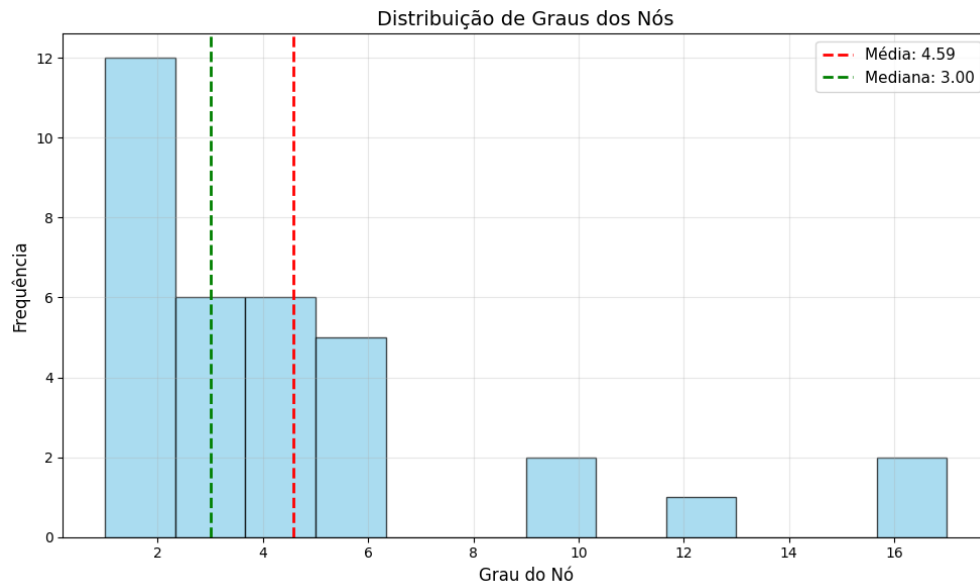


Figure 3.1: Node Degree Distributions

Figure-3.2, titled "Node Degree Distribution", presents a histogram with the number of nodes (frequency) as a function of their respective degrees. The dashed lines indicate the mean (4.59) and median (3.00) of the degrees.

The difference between these two values shows a right-skewed distribution; that is, there are few nodes with a very high degree that raise the mean, but most nodes have low degrees. The greater concentration of bars in the degree ranges between 2 and 5 reinforces the idea that the graph is predominantly composed of nodes with few connections.

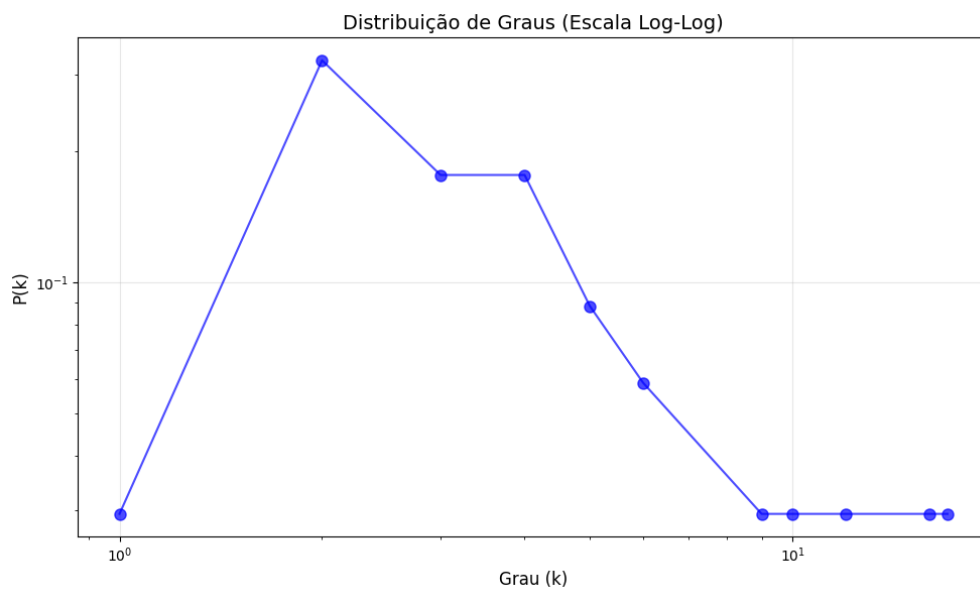


Figure 3.2: Node Degree Distributions in Log

Together, both graphs demonstrate that the Karate Club Graph is not uniform but rather unequal in the distribution of connections, which indicates the presence of a hierarchical and centralized structure. This characteristic explains why certain members are more influential in information propagation or group cohesion.

Furthermore, the long-tail shape observed in the log-log graph suggests that the network presents properties similar to those of a scale-free network, typical of real social systems, in which few individuals exercise leadership and mediation roles between different subgroups.

Identifying Strategic Actors

4.1. Betweenness Centrality

Figure-5.1 presented shows Zachary's Karate Club Graph, colored according to the betweenness centrality of each node. This metric quantifies how much a vertex acts as a bridge between other pairs of vertices; that is, it measures the number of shortest paths that pass through it.

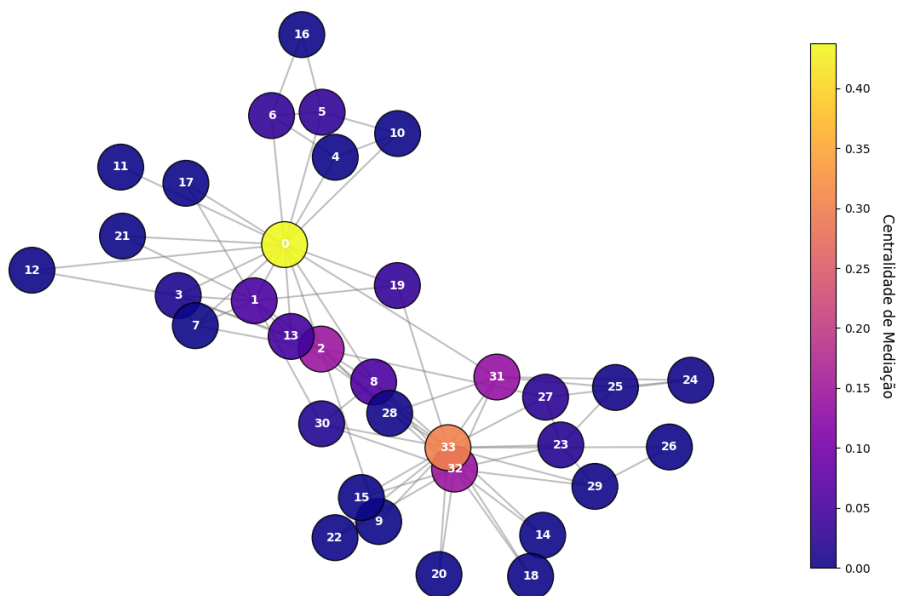


Figure 4.1: Betweenness Centrality

The nodes were colored according to the value of this centrality, as shown in the color bar on the right

side of the figure. Yellowish and orange tones indicate vertices with high betweenness centrality, while purple and dark blue tones correspond to nodes with low betweenness.

In the displayed network, it is observed that nodes 0 and 33 present the most intense colors (yellow and orange), which indicates that they are the most central vertices in the communication flow between club members. This means that they are frequently present in the shortest paths between other pairs of nodes, playing a fundamental role in linking different groups in the network. Historically, these nodes represent the club's instructor and administrative president — precisely the two individuals who led opposing factions during the conflict that led to the group's division.

The remaining vertices, represented in purple and blue tones, have less structural relevance, being more peripheral and having connections restricted to their immediate circle of friends. This configuration demonstrates that the network has two poles of influence and that its cohesion strongly depends on the mediation between these leaders.

The graph shows that betweenness centrality is a powerful metric for identifying strategic actors in a social network, responsible for connecting subgroups and maintaining the flow of information between them. In the case of the Karate Club, the analysis reinforces that nodes 0 and 33 were the main intermediaries and influencers of the social system, and the rupture between them naturally caused the network's fragmentation into two distinct communities.

5

Number of Direct Connections

5.1. Degree Centrality

Figure-5.1 represents Zachary's Karate Club Graph, colored according to the degree centrality metric. This metric evaluates the importance of a vertex based on the number of direct connections it has. In other words, the more links a node has with other nodes, the greater its degree centrality and, consequently, its local influence within the network.

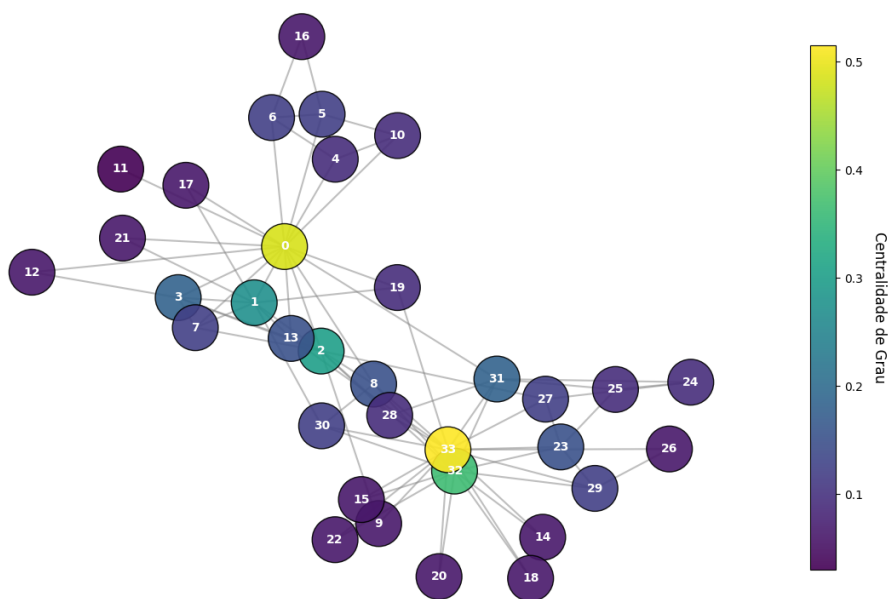


Figure 5.1: Degree Centrality

In the image, nodes are colored according to a color gradient shown on the sidebar: yellowish tones indicate high centrality, while purple tones represent low centrality. This visual encoding allows for the quick identification of which club members (nodes) have more connections with other participants.

It is observed that nodes 0 and 33 stand out with yellow and light green coloring, indicating that they are the vertices with the highest number of direct connections in the network. These two individuals played central roles in the club's social structure: node 0 represents the instructor, and node 33 represents the administrative president. Both were influential figures who attracted several members around them, which later led to the group's division into two communities.

Nodes with intermediate coloring (blue tones) have a moderate number of connections, acting as local intermediaries within small subgroups. Meanwhile, nodes with purplish tones have low degree centrality, being located on the network's peripheries and connecting to few other members.

This distribution indicates that the club's social network is highly unequal in terms of connectivity, with few members concentrating most of the direct relationships, a typical characteristic of real social networks and decentralized power structures, in which some individuals exert a strong influence over others. In summary, the figure demonstrates that the Karate Club Graph presents a structure centered on two poles of high connectivity (nodes 0 and 33), responsible for maintaining cohesion and information flow within the community.

Number of Direct Connections

6.1. Histogram of Node Degree Distribution

Figure 6.1 presents the histogram of the degree distribution of nodes in Zachary's Karate Club Graph, which represents the number of direct connections of each member within the club's social network. On the horizontal axis is the node degree; that is, the number of links a vertex has with other vertices, while the vertical axis indicates frequency, showing how many nodes share the same degree.

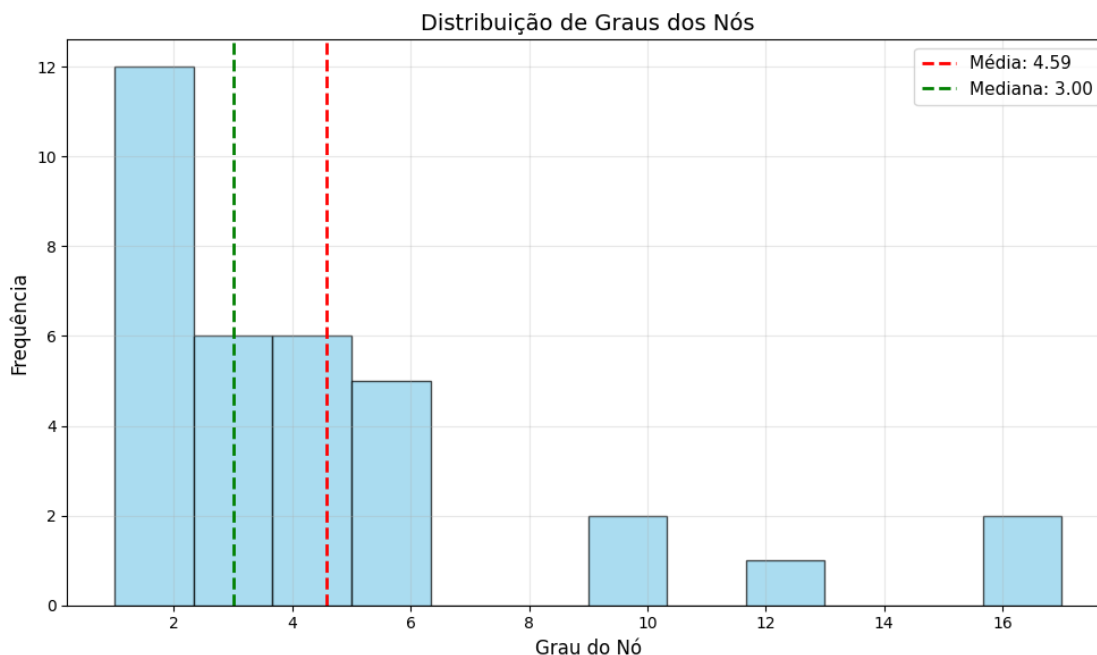


Figure 6.1: Node Degree Distribution

It is observed that most nodes have degrees between 1 and 5, indicating that most club members maintain relationships with few other participants. This behavior is characteristic of real social networks, in which most individuals connect to a reduced circle of contacts.

The dashed lines indicate statistical measures of the distribution: The green line represents the median (3.00), which indicates that half of the nodes have a degree equal to or less than three. The red line marks the mean (4.59), whose value, greater than the median, shows the presence of some nodes with many links, raising the average value.

This difference between the mean and median indicates a right-skewed distribution, typical of non-uniform networks, in which few vertices concentrate most of the connections, the so-called hubs or central nodes.

In summary, the histogram demonstrates that the graph presents a heterogeneous connectivity structure: most members are peripherally connected, while few individuals, such as the instructor and club president, have a high degree and act as centers of influence within the network. This configuration is typical of real social systems, in which some actors play the role of leadership and intermediation between different groups.

Identifying the Intermediaries

7.1. Betweenness Centrality Distribution

Figure-7.1 shows the betweenness centrality distribution of nodes in Zachary's Karate Club Graph. This metric measures how much a node acts as a bridge or intermediary in communication routes between other nodes in the network, that is, how many times a vertex appears in the shortest paths connecting distinct pairs of vertices.

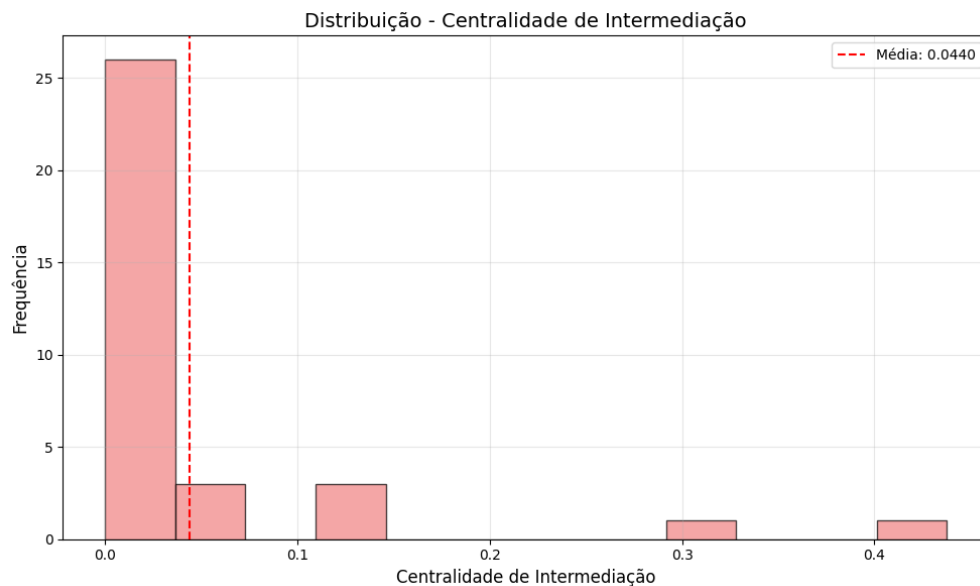


Figure 7.1: Betweenness Centrality Distribution

On the horizontal axis, we have the betweenness centrality values, which range between 0 and approximately 0.45, while the vertical axis indicates the frequency of occurrence of these values among nodes. The red dashed line represents the mean centrality (≈ 0.044), marking the average intermediation value observed in the network.

It is observed that most nodes present very low centrality values, concentrating close to zero, which indicates that most vertices do not play a relevant role as intermediaries in communications between other members. Only a few nodes have high values, forming a long tail to the right of the distribution. These high-centrality nodes are critical for information flow, acting as bridges between different communities or subgroups of the network.

In the context of the Karate Club, these nodes correspond to members who connected different social groups within the club, especially during the period of tension that led to the organization's division. The presence of few vertices with high intermediation reinforces that the network is highly hierarchical and dependent on few mediators, which makes it susceptible to fragmentation if these individuals are removed.

In summary, the graph shows that most members have little influence on the connections between others, while a small number of individuals play a central role in the network's cohesion and global communication.

Hierarchical and Heterogeneous Structures

8.1. Identifying How Much Each Node Is Directly Connected

Figure-8.1 illustrates the degree centrality distribution of nodes in Zachary's Karate Club Graph. This measure quantifies how much each node is directly connected to other vertices in the network, representing the level of local influence of each member within the social system.

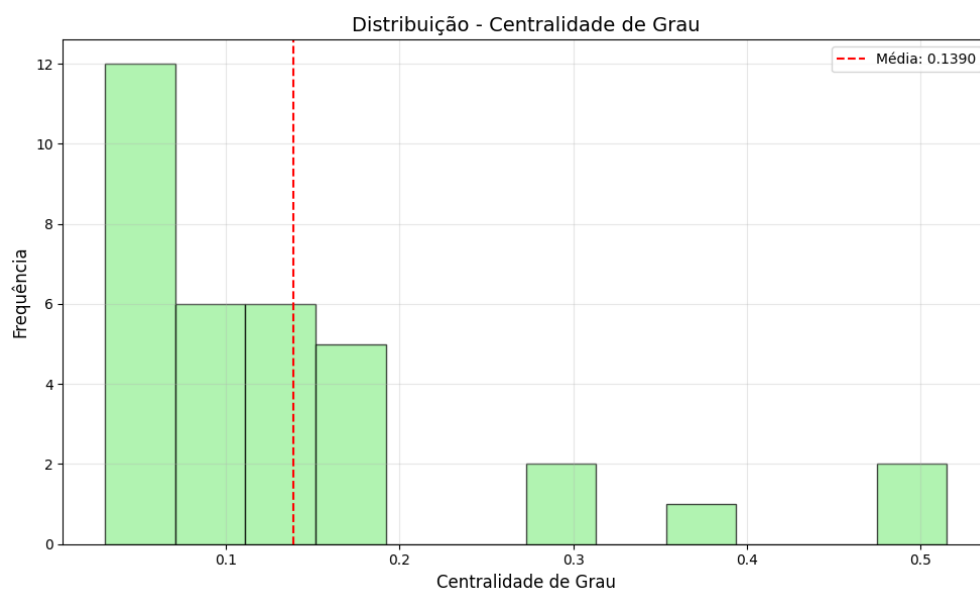


Figure 8.1: Degree Centrality Distribution

On the horizontal axis, degree centrality values are shown, which range from approximately 0.02 to just above 0.5. The vertical axis indicates frequency, that is, how many nodes present a certain centrality interval. The red dashed line marks the mean centrality (≈ 0.139), which serves as a reference to compare the degree of connectivity of nodes.

The shape of the distribution reveals that most nodes have low degree centrality, concentrating in the first intervals on the left side of the graph. This indicates that most club members have few direct connections, interacting only with a small group of other members.

However, there are some nodes that stand out for having significantly higher centrality values, representing the leaders or most influential members of the network, who maintain direct contact with many other individuals.

This characteristic shows a hierarchical and heterogeneous structure, typical of real social networks: few individuals centralize a good part of interactions, while the majority occupy peripheral positions. In practical terms, this means that these central nodes exert a strong influence on the network's communication and cohesion dynamics, being able to disseminate information quickly or connect subgroups that would otherwise be isolated.

The graph demonstrates that the club's network is non-uniform and centralized, with connectivity concentrated in a few highly relevant nodes, a common pattern in natural social and organizational structures.

Efficiency of a Node in Reaching Others

9.1. Closeness Centrality Distribution

Figure-9.1 presents the closeness centrality distribution of nodes in Zachary's Karate Club Graph, a metric that measures how close each vertex is to all others in the network. In other words, closeness centrality indicates the efficiency of a node in reaching others, being inversely proportional to the sum of distances between that node and all others.

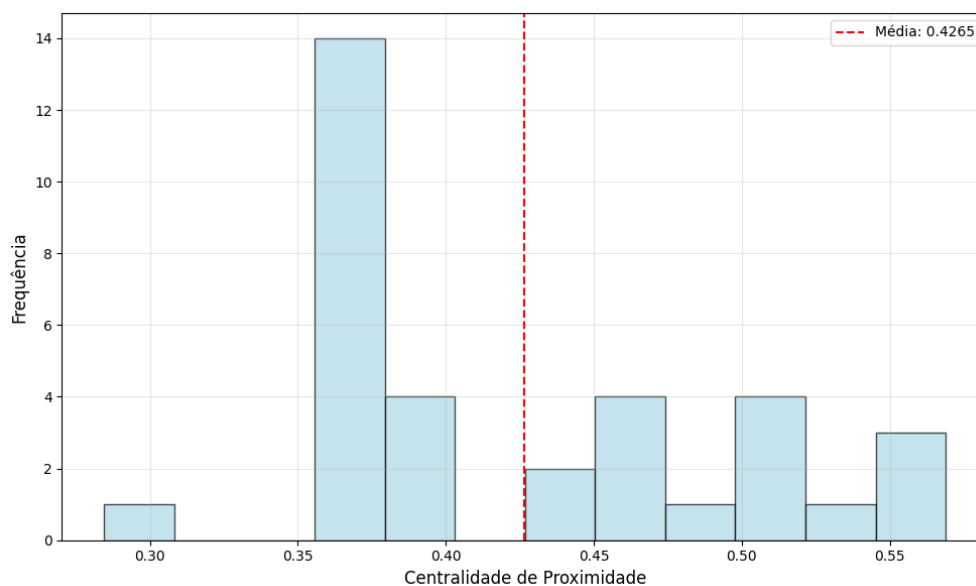


Figure 9.1: Closeness Centrality Distribution

On the horizontal axis, closeness centrality values are displayed, while the vertical axis represents frequency, that is, how many nodes have a certain level of average proximity in relation to others. The red

dashed line indicates the global mean centrality value (≈ 0.4265), serving as a reference for comparison.

It is observed that most nodes present values close to the mean, concentrating between 0.35 and 0.45, which suggests a moderately connected network in which most members can reach others with a relatively small number of interactions. However, there are some nodes with higher values (above 0.5) that represent central vertices, that is, those that are better positioned topologically to spread information quickly and efficiently.

These nodes with high closeness centrality correspond to individuals who can access or influence the network more quickly, playing an important role in processes such as information dissemination, activity coordination, or even conflict mediation within the club.

In contrast, nodes with lower centrality (around 0.3) occupy peripheral positions, being more distant from others; therefore, they are less efficient in terms of communication or influence.

In summary, the graph shows that the network presents an asymmetric structure in which few nodes have quick access to all others, while the majority occupy median or peripheral positions. This configuration reflects a hierarchical social organization, with the presence of strategically positioned individuals to ensure cohesion and the efficient flow of information within the group.

10

Identifying Structural Leaderships in the Network

10.1. Degree Versus Closeness

Figure-10.1 shows the relationship between node degree and betweenness centrality, colored according to closeness centrality, for Zachary's Karate Club Graph. Each point represents a node (or network member), and the position, color, and height reflect different aspects of its structural importance within the social network.

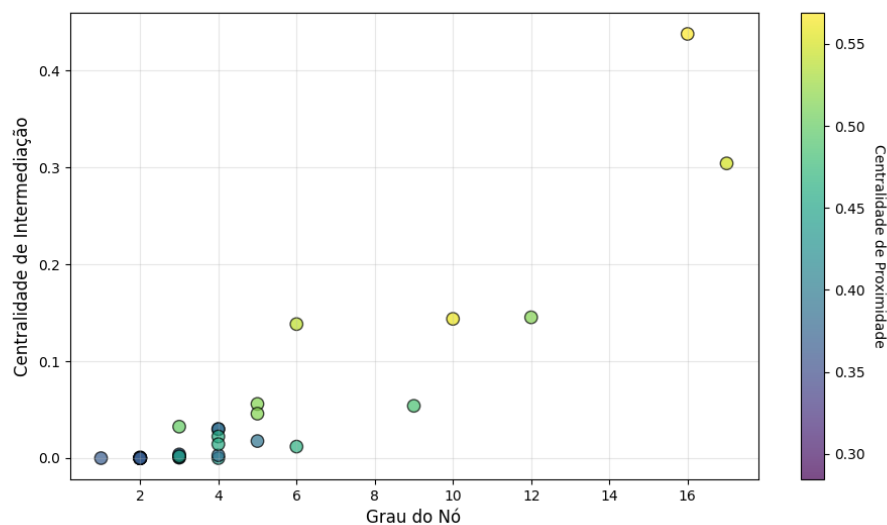


Figure 10.1: Degree Versus Closeness

On the horizontal axis, we have the node degree; that is, the number of direct connections it maintains with other nodes. The vertical axis represents betweenness centrality, which indicates the frequency with which a node acts as a bridge between others; the higher the value, the more the node is responsible for mediating communications in the network. The coloring of points, from blue to yellow, indicates closeness centrality, showing how quickly a node can reach other members.

Joint analysis reveals a characteristic pattern of real social networks: most nodes have low degree and low betweenness, situated in the lower left part of the graph; these are peripheral nodes with little structural influence. In contrast, few nodes exhibit high simultaneous values of degree and betweenness, located more to the right and in the upper part of the graph.

These nodes are highly connected and strategic, functioning as hubs and network mediators; they not only have many direct connections but also control the flow of information between different groups.

The lighter (yellowish) coloring in nodes with higher betweenness degrees reinforces that these vertices also have high closeness centrality, meaning they are topologically well-positioned and can communicate efficiently with all other network members.

In summary, the graph shows a strong correlation between the three centrality measures: nodes with high degree tend to present higher betweenness and closeness values. This demonstrates the existence of structural leadership in the network, with few individuals playing an essential role in cohesion, information dissemination, and the group's global connectivity.

11

Structural Analysis of Communities in Networks

11.1. Community Detection

Figure 11.1 presents the community detection in Zachary's Karate Club graph, a classic example of a social network in which nodes represent club members and edges correspond to interactions or friendship ties between them.

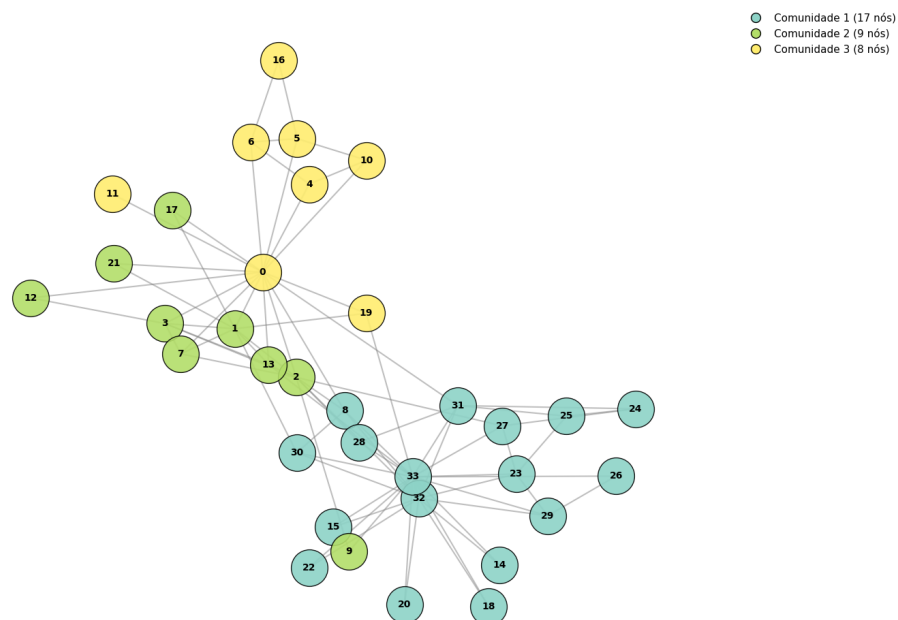


Figure 11.1: Community Detection

The algorithm employed for segmentation identifies cohesive groups of nodes, called communities, which have a greater density of internal connections than external to the community. In the figure, each color represents a distinct community:

- Community 1 (light blue, 17 nodes): located mainly in the lower and right part of the graph, it is the largest identified group. It presents a dense structure and significant connections among its members, with a well-defined core centered on nodes 32 and 33.
- Community 2 (green, 9 nodes): occupies the central-left region of the graph. It acts as a structural bridge between other communities, as evidenced by nodes 1, 2, and 13, which connect Community 1 to Community 3.
- Community 3 (yellow, 8 nodes): situated in the upper part of the graph, it represents a more compact and homogeneous group, centered on node 0, which has a high degree of connectivity and betweenness, functioning as the local leader of this sub-network.

The modular structure of the network shows the existence of well-defined subgroups, a common characteristic in real social networks. This reflects the natural behavior of alliance and affinity formation among individuals, in which those with higher interaction frequency tend to form cohesive blocks.

Furthermore, the nodes that interconnect different communities, such as 2, 13, and 33, are of strategic importance, as they act as communication links between distinct groups. These vertices play an essential role in the network's global cohesion, preventing its fragmentation.

In summary, the graph reveals the presence of three main communities, with distinct roles:

1. An upper community is cohesive around the original leader (node 0);
2. A lower community is organized around node 33;
3. An intermediate community connects both, representing a transition between two poles of influence.

This structure reflects the karate club's own historical conflict, in which there was a division between the instructor and the administrator, a phenomenon documented in Zachary's original study (1977).

12

Matrix Representation of Communities

12.1. Graph Adjacency Matrix (Ordered by Communities)

Figure 12.1 presents the adjacency matrix of Zachary's Karate Club graph, organized according to detected communities. Each cell represents the presence (1) or absence (0) of a link between two nodes. Darker shades indicate a higher concentration of connections.

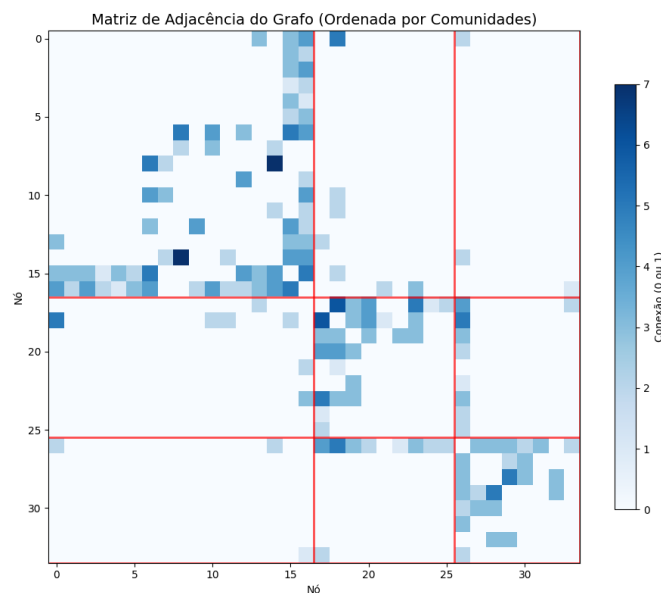


Figure 12.1: Graph Adjacency Matrix (Ordered by Communities)

The ordering of nodes according to communities reveals well-defined structural blocks along the main diagonal, reflecting the high density of connections within each group. These blocks represent intra-community connections, with more intense coloring, while lighter regions outside the diagonal indicate inter-community links, which are rarer and more dispersed.

The red lines visually delimit the boundaries between communities, highlighting the network's modular segmentation. This configuration confirms that the graph presents a strongly modular structure, characterized by cohesive and relatively independent subgroups, a recurring pattern in real social networks, where individuals tend to organize themselves in circles of affinity and frequent interaction.

13

Calculation and Interpretation of Local Clustering

13.1. Clustering Coefficient

Figure 13.1 presents the local clustering coefficient for each node of Zachary's Karate Club graph. This metric measures the degree of cohesion among a vertex's neighbors, indicating how much its contacts also connect with one another.

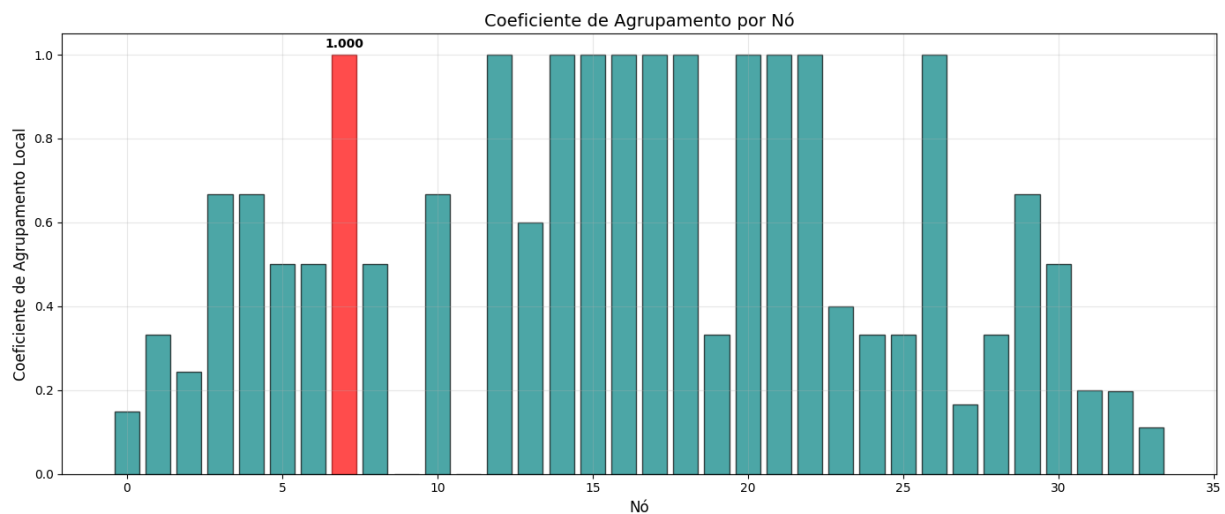


Figure 13.1: Clustering Coefficient

Values close to 1 represent nodes whose neighbors form a complete subgraph (clique), while values close to 0 indicate the absence of connections between neighbors.

The highlighted node in red (node 6) presents a maximum coefficient ($C = 1,000$), demonstrating complete interconnectivity among its neighbors. Other nodes also exhibit high C values, located mainly at the edges of detected communities, which reinforces the local cohesion of these regions.

On the other hand, nodes with a low clustering coefficient, such as nodes 0 and 33, play a structural bridge role, connecting distinct communities and ensuring the network's global integrity.

These results indicate that the network combines densely connected areas with strategic linking nodes, reflecting the typical social structure of real communities, in which small cohesive groups are interconnected by individuals with an intermediation role.

14

Betweenness Centrality

14.1. Top 10 Nodes by Betweenness Centrality

Figure 14.1 presents the ten nodes with the highest betweenness centrality values in Zachary's Karate Club graph. This metric evaluates the importance of a vertex as a communication link between different parts of the network, and it being calculated based on the number of shortest paths that pass through it.

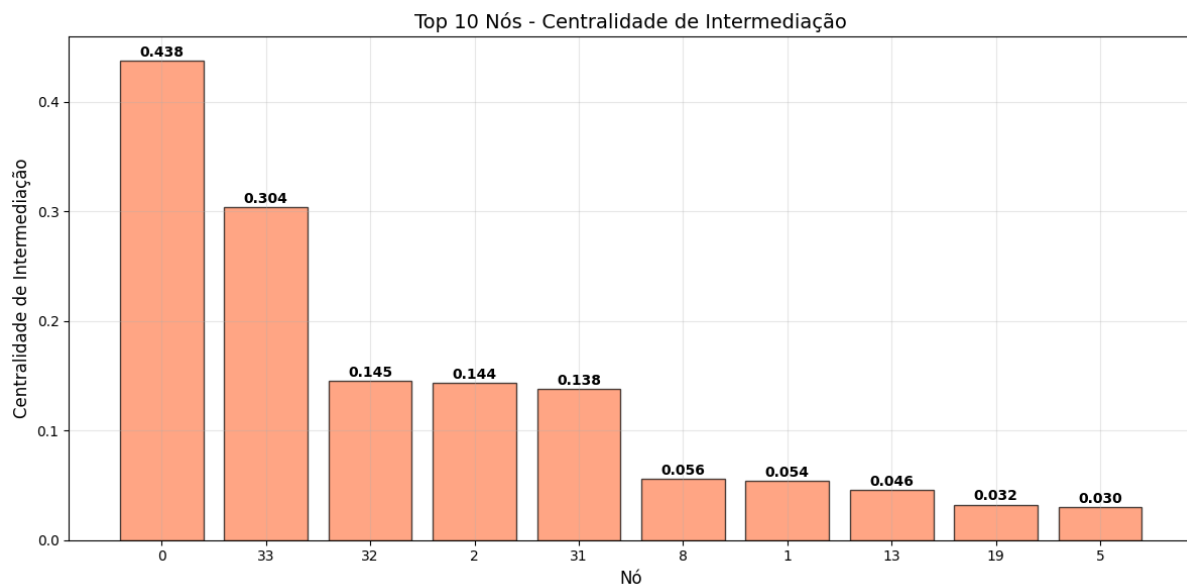


Figure 14.1: Top 10 Nodes by Betweenness Centrality

Nodes 0 and 33 stand out expressively, with values of 0,438 and 0,304, respectively. These nodes represent the two club leaders: the administrator and the instructor, and they are responsible for connecting the network's main communities. Their role is crucial in maintaining global connectivity and disseminating information between distinct groups.

Nodes 2, 31, and 32 present intermediate centralities, functioning as local bridges within communities. Meanwhile, nodes 1, 5, 8, 13, and 19 exhibit smaller values, with action more restricted to internal connections.

This distribution reveals a highly centralized structure, characteristic of social networks in which few individuals concentrate control over communication flow. The removal of these nodes with higher centrality could cause significant fragmentation in the network, evidencing their strategic role in the structural cohesion of the system.

15

Analysis of Hubs by Degree Centrality

15.1. Top 10 Nodes by Degree Centrality

Figure 15.1 presents the ten nodes with the highest degree centrality values in Zachary's Karate Club graph. This metric represents the fraction of nodes directly connected to a vertex and reflects its local importance in the network structure.

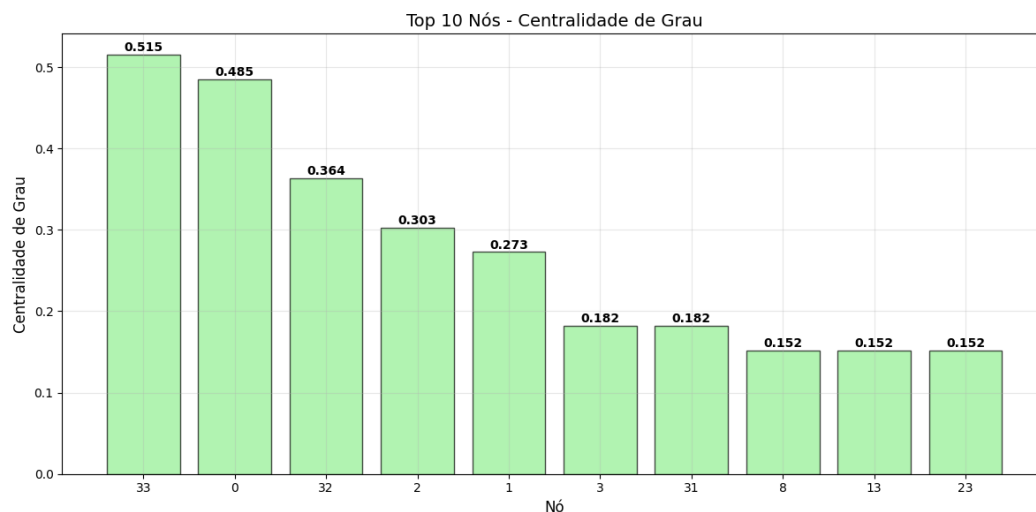


Figure 15.1: Top 10 Nodes by Degree Centrality

Nodes 33 and 0 have the highest centralities, with values of 0,515 and 0,485, respectively. These vertices correspond to the two network leaders: the instructor and the administrator, who play a fundamental role in the internal communication of their communities.

Nodes 32, 2, and 1 present intermediate values (between 0,36 and 0,27), representing relevant actors in their sub-networks, while nodes 3, 31, 8, 13, and 23, with values between 0,18 and 0,15, have more restricted influence.

This distribution confirms the heterogeneous nature of the graph: few vertices concentrate most of the connections, configuring a network with hub characteristics, which are essential elements for structural cohesion and efficient information propagation. The presence of these highly connected nodes makes the network robust to random failures but vulnerable to the selective removal of these central vertices.

16

Number of Direct Connections

16.1. Top 10 Nodes by Closeness Centrality

Figure 16.1 presents the ten nodes with the highest closeness centrality values in Zachary's Karate Club graph. This metric indicates how close a vertex is to the others in the network, representing its efficiency in information dissemination and access to different parts of the system.

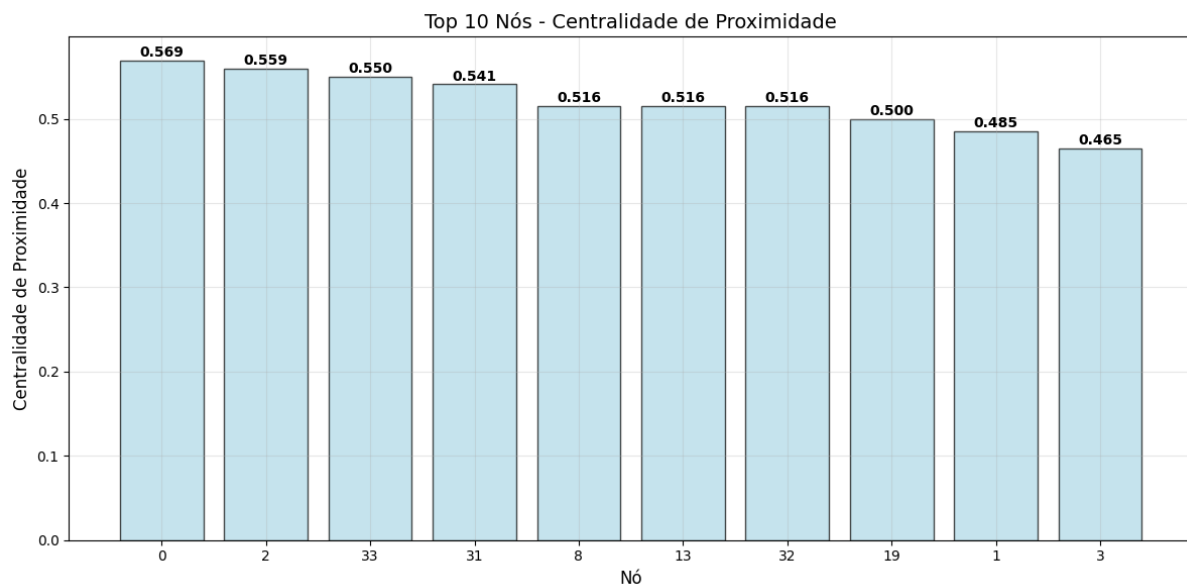


Figure 16.1: Top 10 Nodes by Closeness Centrality

Nodes 0, 2, and 33 stand out with the highest centrality values (0,569, 0,559, and 0,550, respectively), which characterize them as central elements in the network's global communication. These vertices function as structural links between the main communities, allowing short paths between distinct groups.

Nodes 31, 8, 13, and 32, with values close to 0,516, occupy intermediate positions, acting as facilitators of integration between subgroups. Meanwhile, nodes 19, 1, and 3, with values between 0,50 and 0,46, present more local influence.

Together, these results show that the network has an efficient and cohesive structure in which the average distance between vertices is small, favoring fast communication and constant interaction among members.

17

Number of Direct Connections

17.1. Correlation Matrix

Figure-17.1 presents the correlation matrix between the main centrality metrics and the clustering coefficient of Zachary's Karate Club graph. This matrix shows how the different measures of importance and the local structure of vertices relate to each other, allowing us to identify redundancies, complementarities, and contrasts between the indicators used in network analysis.

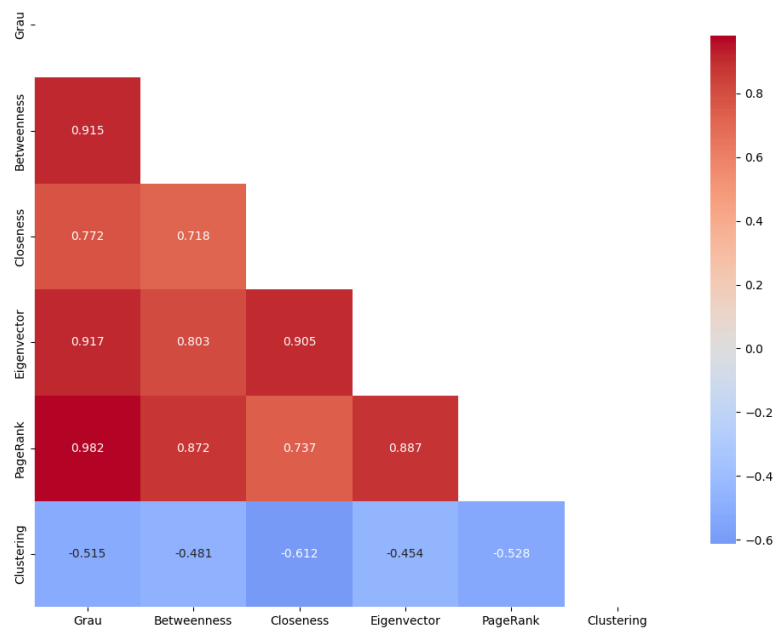


Figure 17.1: Correlation Matrix

A strong positive correlation is observed between centrality metrics, especially between degree, PageRank, betweenness, and eigenvector, with values above 0.87. This indicates that nodes with the highest number of direct connections (high degree) also tend to be important in terms of intermediation and global influence, reinforcing the presence of structural hubs. The Degree-PageRank correlation (0.982) is particularly high, showing that popularity (degree) and authority (PageRank) are strongly aligned in this graph.

Closeness centrality presents intermediate correlations with other measures (between 0.71 and 0.77), suggesting that it captures a complementary dimension: the efficiency of global access, more than the number or quality of local connections.

On the other hand, the clustering coefficient shows a negative correlation with all centrality metrics, varying between -0.45 and -0.61 . This behavior is expected: nodes with high centrality, generally hubs, tend to connect with multiple groups that do not necessarily connect with each other, thereby reducing the density of local triangles.

In summary, the matrix confirms that the graph has a hierarchical and non-homogeneous structure, in which few highly connected nodes play a global mediation role, while most vertices exhibit more local connections and a high level of intra-community cohesion.